

Lesson 01: LocalStack Setup

Learning Objectives

- Understand what LocalStack is and why it exists
 - Install and run LocalStack via Docker
 - Verify the installation is working
 - Explore the LocalStack Web UI
 - Understand the service support matrix (free vs pro)
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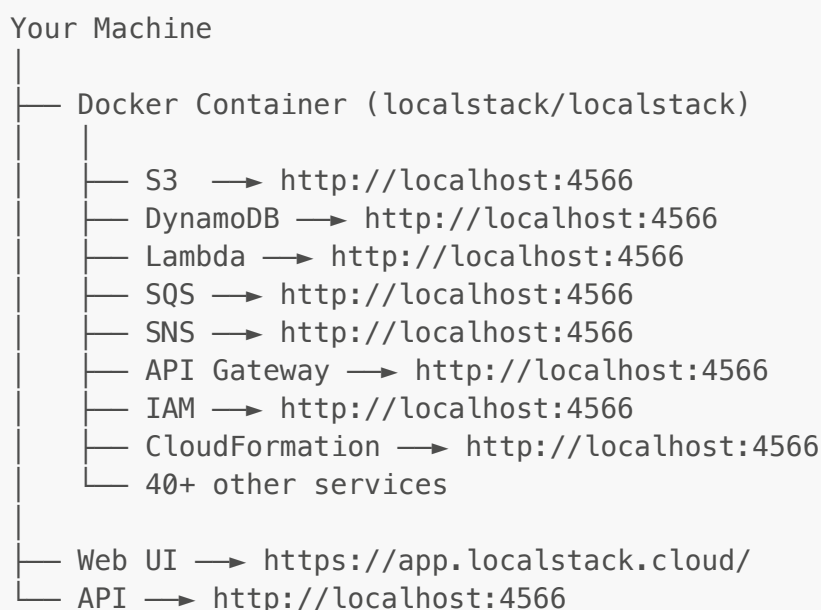
1. What is LocalStack?

LocalStack is a fully functional local AWS cloud stack. It emulates AWS services on your local machine so you can develop and test cloud applications **without an internet connection** and **without incurring AWS costs**.

Why use LocalStack for learning Terraform?

- **No AWS account needed** — zero cost, no credit card required
 - **Fast feedback loop** — resources create in milliseconds
 - **Safe experimentation** — no risk of accidental production charges
 - **Offline capable** — develop anywhere
 - **Real AWS compatibility** — skills transfer directly to production
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2. LocalStack Architecture



All AWS services are accessed through a **single endpoint**:

```
http://localhost:4566
```

3. Installation

Note on LocalStack versions: As of 2025, the `localstack/localstack:latest` Docker image defaults to the **Pro** edition, which requires an auth token. The free **Community** edition (`:3.0.0` or later pinned tags) does not require a token but still benefits from having one for Web UI access.

All lessons in this curriculum use only **free/Community features**. You have two installation options below.

Prerequisites

- Docker installed on your machine
- Docker daemon running

Verify Docker:

```
docker --version
docker info
```

Option 1: LocalStack Desktop App (Recommended for Beginners)

LocalStack provides a native desktop application with a built-in Docker manager, Web UI launcher, and one-click startup. This is the easiest way to get started.

1. Go to <https://docs.localstack.cloud/getting-started/installation/#desktop-app>
2. Download the installer for your OS:
 - **macOS:** `.dmg` file
 - **Windows:** `.exe` installer
 - **Linux:** `.AppImage` or `.deb/.rpm`
3. Install and launch the app
4. On first launch:
 - You'll be prompted to sign in to your LocalStack account (create one if you haven't already)
 - Go to **Settings → Auth Token** and paste your token (see "Get a LocalStack Auth Token" below)
5. Click **"Start LocalStack"** — the app handles Docker pulling, port mapping, and startup automatically
6. The Web UI opens at `http://localhost.localstack.cloud:4566` once ready

The Desktop App is ideal for visual learners. You can always switch to the Docker CLI approach later.

Option 2: Official LocalStack CLI Installation

Follow the official CLI installation guide:

<https://docs.localstack.cloud/getting-started/installation/>

This covers `pip install localstack`, Homebrew on macOS, and other package managers.

Option 3: Docker Installation (Most Common)

Option 3: Docker Installation

Step 1: Get a LocalStack Auth Token

LocalStack uses auth tokens to unlock the Web UI and Pro features. The free tier is available with a token:

- 1. Go to <https://app.localstack.cloud>
- 2. Sign up for a free account (email or GitHub login)
- 3. After logging in, navigate to your **Auth Tokens** page:
 - Click your profile icon (top-right)
 - Select **"Auth Tokens"** from the dropdown
- 4. Click **"Generate Token"**
- 5. Copy the token — it looks like: `ls-xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx`

Keep this token handy. You'll use it in the `docker run` command as `LOCALSTACK_AUTH_TOKEN`.

Step 2: Run LocalStack

```
docker run \
-d \
--name localstack \
-p 4566:4566 \
-p 4510-4559:4510-4559 \
-e LOCALSTACK_AUTH_TOKEN=<your-token> \
localstack/localstack
```

Flag explanation:

Flag	Purpose
<code>-d</code>	Detached mode (runs in background)
<code>--name localstack</code>	Container name for easy reference
<code>-p 4566:4566</code>	Main API endpoint
<code>-p 4510-4559:4510-4559</code>	Service-specific endpoints (Lambda, etc.)
<code>-e LOCALSTACK_AUTH_TOKEN</code>	Your auth token from Step 1

Verify the Container is Running

```
docker ps
```

Expected output includes a line with `localstack/localstack` and status `Up`.

You can also check logs:

```
docker logs localstack
```

4. Verify the Installation

Health Check

```
curl -s http://localhost:4566/_localstack/health | python3 -m json.tool
```

Expected JSON response shows all services with `"available"` status:

```
{
  "services": {
    "s3": "available",
    "dynamodb": "available",
    "sqs": "available",
    "sns": "available",
    "lambda": "available",
    "iam": "available",
    "apigateway": "available",
    "cloudformation": "available",
    ...
  }
}
```

Check the Web UI

Open in your browser:

```
http://localhost.localstack.cloud:4566
```

You should see the LocalStack Web UI with a navigation panel listing available services.

5. Exploring the Web UI

Navigate through these services in the UI to familiarize yourself:

Service	What to look for
S3	Bucket listing, create bucket button
Lambda	Function listing, create function form
DynamoDB	Table listing, create table form
SQS	Queue listing, create queue form
SNS	Topic listing, create topic form
IAM	Users, roles, policies
CloudFormation	Stack management

Observation exercise: Try creating resources manually from the UI:

1. Create an S3 bucket named `lesson-01-bucket`
2. Create a DynamoDB table named `LessonTable` with a primary key `id` (String)
3. Create an SQS queue named `lesson-queue`

Note the ARN format and resource IDs that LocalStack auto-generates.

6. Service Support Matrix

LocalStack offers different tiers. For this curriculum, the **free/community edition** is sufficient.

Free Tier (Community)

Fully Supported	Notes
S3	Object storage
DynamoDB	NoSQL database
SQS	Message queues
SNS	Pub/sub notifications
Lambda	Serverless functions
IAM	Identity and access
CloudFormation	Infrastructure as code
API Gateway	REST APIs
CloudWatch Logs	Log management (partial)
SSM	Parameter store
Secrets Manager	Secrets storage

Fully Supported	Notes
Route53	DNS (basic)
KMS	Key management (basic)

Pro Tier Only (not required)

- ECS / EKS (full support)
- CloudWatch Metrics (full)
- WAF
- CloudFront
- Cognito

Unsupported (even in Pro)

- Some very recent service additions
- Services requiring physical hardware (AWS Ground Station, etc.)

7. Stopping and Restarting LocalStack

```
# Stop
docker stop localstack

# Start again
docker start localstack

# Remove and recreate fresh
docker rm -f localstack
docker run -d --name localstack -p 4566:4566 -p 4510-4559:4510-4559 -e
LOCALSTACK_AUTH_TOKEN=<your-token> localstack/localstack
```

Important: Stopping the container **preserves** your data. Removing the container **destroys** all resources.

8. Key Takeaways

- LocalStack runs in Docker on a single port (4566)
- All AWS services are accessed through one endpoint
- The Web UI provides visual resource management
- Free tier covers all services needed for this curriculum
- Data persists as long as the container exists (no removal)

9. Troubleshooting

Problem	Solution
<code>docker: command not found</code>	Install Docker Desktop
Port 4566 already in use	Stop other services: <code>lsof -i :4566</code>
Container exits immediately	Check Docker resources (RAM/CPU)
Web UI not loading	Use <code>http://localhost:4566</code> instead of <code>localhost.localstack.cloud</code>
Health check returns empty	Wait 10-20 seconds for startup

example and exercise repo url

<https://github.com/eyu-se/terraform-training-material-with-projects>

Trainer - Eyuel M.